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[Paper 58 continued]

NOTES ON THE ABELIAN INTEGRALS.—JACOBI'S SYSTEM OF DIFFERENTIAL EQUATIONS.

each of which contains only a single root of the equation $fx = 0$. But in Richelot's second memoir "Einige neue Integralgleichungen des Jacobischen Systems Differential- gleichungen" (*Crelle*, t. xxv. [1843] p. 97), the equations are obtained by direct inte- gration in a form not involving any of the roots of this equation; the method employed in obtaining them being in a great measure founded upon the memoir just quoted of Jacobi's. The following is the process of integration.

Denoting the variables by x_1, $x_2, \ldots x_n$, and writing

$$Fa = (a - x_1)(a - x_2) \ldots (a - x_n),$$

so that

$$F'x_1 = (x_1 - x_2) \ldots (x_1 - x_n),$$

&c.

then the system of differential equations is satisfied by assuming that x_1, $x_2, \ldots x_n$ are functions of a new variable t, determined by the equations

$$\frac{dx_1}{dt} = \frac{\sqrt{(fx_1)}}{F'x_1}, \quad \&c.$$

(In fact these equations give $\Sigma \dfrac{dx}{\sqrt{(fx)}} = dt\, \Sigma \dfrac{1}{F'x} = 0$, &c.)

From these we deduce, by differentiation,

$$\frac{d^2x_1}{dt^2} = \frac{1}{2} \frac{d}{dx_1} \frac{fx_1}{(F'x_1)^2} + \frac{\sqrt{(fx_1)}}{F'x_1} \Sigma' \frac{\sqrt{(fx)}}{(x_1 - x)F'x}$$

(where Σ' refers to all the roots except x_1) and a set of analogous equations for x_2, $x_3, \ldots x_n$.

Dividing this by $a - x_1$, where a is arbitrary, and reducing by

$$\frac{1}{(a - x_1)(x_1 - x)} = \frac{1}{2(a - x)(a - x_1)} \left(1 - \frac{x + x_1 - 2a}{x_1 - x}\right),$$

we have

$$\frac{1}{a - x_1} \frac{d^2x_1}{dt^2} = \frac{1}{2(a - x_1)} \frac{d}{dx_1} \frac{fx_1}{(F'x_1)^2} + \frac{1}{2} \frac{\sqrt{(fx_1)}}{(a - x_1)F'x_1} \Sigma' \frac{\sqrt{(fx)}}{(a - x)F'x}$$
$$- \frac{1}{2} \Sigma' \frac{\sqrt{(fx)}\sqrt{(fx_1)}}{F'x\, F'x_1} \frac{x_1 + x - 2a}{(a - x)(a - x_1)(x_1 - x)},$$

that is

$$\frac{1}{a - x_1} \frac{d^2x_1}{dt^2} = \frac{1}{2(a - x_1)} \frac{d}{dx_1} \frac{fx_1}{(F'x_1)^2} + \frac{1}{2} \frac{\sqrt{(fx_1)}}{(a - x_1)F'x_1} \Sigma \frac{\sqrt{(fx)}}{(a - x)F'x}$$
$$- \frac{1}{2} \frac{fx_1}{(a - x_1)^2 (F'x_1)^2} - \frac{1}{2} \Sigma' \frac{\sqrt{(fx)}\sqrt{(fx_1)}}{F'x\, F'x_1} \frac{x_1 + x - 2a}{(a - x)(a - x_1)(x_1 - x)}.$$

and taking the sum of all the equations of this form, the last term disappears on account of the factor $x_1 - x$ in the denominator, and the result is

$$\Sigma \frac{1}{a-x}\frac{d^2x}{dt^2} = \frac{1}{2}\Sigma \frac{1}{a-x}\frac{d}{dx}\frac{fx}{(F'x)^2} + \frac{1}{2}\left\{\Sigma \frac{\sqrt{(fx)}}{(a-x)F'x}\right\}^2 - \frac{1}{2}\Sigma \frac{fx}{(a-x)^2(F'x)^2}.$$

This being premised, assume

$$y = \sqrt{(Fa)},$$

which, by differentiation, gives

$$\frac{dy}{dt} = -\frac{1}{2}y\Sigma \frac{1}{a-x}\frac{dx}{dt} = -\frac{1}{2}y\Sigma \frac{\sqrt{(fx)}}{(a-x)F'x},$$

and thence

$$\frac{d^2y}{dt^2} = -\frac{1}{2}\frac{dy}{dt}\Sigma \frac{1}{a-x}\frac{dx}{dt} + \frac{1}{2}y\Sigma \frac{1}{(a-x)^2}\left(\frac{dx}{dt}\right)^2 - \frac{1}{2}y\Sigma \frac{1}{a-x}\frac{d^2x}{dt^2},$$

that is

$$\frac{d^2y}{dt^2} = \frac{1}{4}y\left\{\Sigma \frac{\sqrt{(fx)}}{(a-x)F'x}\right\}^2 - \frac{1}{2}y\Sigma \frac{fx}{(a-x)^2(F'x)^2} - \frac{1}{2}y\Sigma \frac{1}{a-x}\frac{d^2x}{dt^2}.$$

Substituting the preceding value of

$$\Sigma \frac{1}{a-x}\frac{d^2x}{dt^2},$$

we have

$$\frac{d^2y}{dt^2} = -\frac{1}{4}y\Sigma \frac{fx}{(a-x)^2(F'x)^2} - \frac{1}{4}y\Sigma \frac{1}{a-x}\frac{d}{dx}\frac{fx}{(F'x)^2};$$

that is

$$4\frac{d^2y}{dt^2} + y\left\{\Sigma \frac{fx}{(a-x)^2(F'x)^2} + \Sigma \frac{1}{a-x}\frac{d}{dx}\frac{fx}{(F'x)^2}\right\} = 0.$$

Now the fractional part of $\dfrac{fa}{(Fa)^2}$ is equal to

$$\Sigma \frac{fx}{(a-x)^2(F'x)^2} + \Sigma \frac{1}{a-x}\frac{d}{dx}\frac{fx}{(F'x)^2}.$$

Also if L be the coefficient of a^{2n} in fa, the integral part is simply equal to L, (since $(Fa)^2$ is a function of the order $2n$, in which the coefficient of a^{2n} is unity). Hence the coefficient of y in the last equation is simply

$$\frac{fa}{(Fa)^2} - L, \quad = \frac{fa}{y^4} - L;$$

or we have

$$4\frac{d^2y}{dt^2} + y\left(\frac{fa}{y^4} - L\right) = 0,$$

viz. multiplying by the factor $2\dfrac{dy}{dt}$, and integrating,

$$4\left(\frac{dy}{dt}\right)^2 - \frac{fa}{y^2} - Ly^2 = C.$$

Hence replacing y and $\dfrac{dy}{dt}$ by their values

$$\sqrt{(Fa)} \quad \text{and} \quad -\frac{1}{2}\sqrt{(Fa)}\Sigma \frac{\sqrt{(fx)}}{(a-x)F'x},$$

we have

$$Fa\left\{\Sigma\frac{\sqrt{(fx)}}{(a-x)F'x}\right\}^2-\frac{fa}{Fa}-LFa=C,$$

for one of the integrals of the proposed system of equations: and since a is arbitrary, the complete system is obtained by giving any $(n-1)$ particular values to a, and changing the value of the constant of integration C; or by expanding the first side of the equation in terms of a, and equating the different coefficients to arbitrary constants. The *a posteriori* demonstration that all the results so obtained are equivalent to $(n-1)$ independent equations would probably be of considerable interest.

59.

ON THE THEORY OF ELIMINATION.

[From the Cambridge and Dublin Mathematical Journal, vol. iii. (1848), pp. 116–120.]

Suppose the variables X_1, $X_2,\ldots$, g in number, are connected by the h linear equations

$$\Theta_1=a_1X_1+a_2X_2\ldots=0,$$

$$\Theta_2=\beta_1X_1+\beta_2X_2\ldots=0,$$

$$\vdots$$

these equations not being all independent, but connected by the k linear equations

$$\Phi_1=a_1'\Theta_1+a_2'\Theta_2+\ldots=0,$$

$$\Phi_2=\beta_1'\Theta_1+\beta_2'\Theta_2+\ldots=0,$$

$$\vdots$$

these last equations not being independent, but connected by the l linear equations

$$\Psi_1=a_1''\Phi_1+a_2''\Phi_2+\ldots=0,$$

$$\Psi_2=\beta_1''\Phi_1+\beta_2''\Phi_2+\ldots=0,$$

$$\vdots$$

and so on for any number of systems of equations.

Suppose also that $g-h+k-l+\ldots=0$; in which case the number of quantities X_1, $X_2,\ldots$ will be equal to the number of really independent equations connecting them, and we may obtain by the elimination of these quantities a result $\nabla=0$.

To explain the formation of this final result, write

$$\nabla=\left|\begin{array}{llllll}
a_1, & \beta_1\ldots & & & & \\
a_2, & \beta_2\ldots & & & & \\
\vdots & & & & & \\
\hline
a_1', & a_2'\ldots & a_1'', & \beta_1''\ldots & & \\
\beta_1', & \beta_2'\ldots & a_2'', & \beta_2''\ldots & & \\
\vdots & & \vdots & & & \\
\hline
 & & a_1''', & a_2'''\ldots & & \\
 & & \beta_1''', & \beta_2'''\ldots & & \ddots \\
 & & \vdots & & &
\end{array}\right|$$

which for shortness may be thus represented,

$$\nabla = \begin{vmatrix} \Omega & & & \\ \Omega' & \Omega'' & & \\ & \Omega''' & \Omega'''' & \\ & & \ddots & \end{vmatrix}$$

where Ω, Ω', Ω'', Ω''', Ω'''', ... contain respectively h, h, l, l, n, n, ... vertical rows, and g, k, k, m, m, p, ... horizontal rows.

It is obvious, from the form in which these systems have been arranged, what is meant by speaking of a certain number of the vertical rows of Ω' and the supplementary vertical rows of Ω; or of a certain number of the horizontal rows of Ω'' and the supplementary horizontal rows of Ω', &c.

Suppose that there is only one set of equations, or $g = h$: we have here only a single system Ω, which contains h vertical and h horizontal rows, and ∇ is simply the determinant formed with the system of quantities Ω. We may write in this case $\nabla = Q$.

Suppose that there are two sets of equations, or $g = h - k$: we have here two systems Ω, Ω', of which Ω contains h vertical and $h - k$ horizontal rows, Ω' contains h vertical and k horizontal rows. From any k of the h vertical rows of Ω' form a determinant, and call this Q'; from the supplementary $h - k$ vertical rows of Ω form a determinant, and call this Q: then Q' divides Q, and we have $\nabla = Q \div Q'$.

Suppose that there are three sets of equations, or $g = h - k + l$: we have here three systems, Ω, Ω', Ω'', of which Ω contains h vertical and $h - k + l$ horizontal rows, Ω' contains h vertical and k horizontal rows, Ω'' contains l vertical and k horizontal rows. From any l of the k horizontal rows of Ω'' form a determinant, and call this Q''; from the $k-l$ supplementary horizontal rows of Ω', choosing the vertical rows at pleasure, form a determinant, and call this Q'; from the $h-k+l$ supplementary vertical rows of Ω form a determinant, and call this Q: then Q'' divides Q', this quotient divides Q, and we have $\nabla = Q \div (Q' \div Q'')$.

Suppose that there are four sets of equations, or $g = h - k + l - m$: we have here four systems, Ω, Ω', Ω'', and Ω''', of which Ω contains h vertical and $h-k+l-m$ horizontal rows, Ω' contains h vertical and k horizontal rows, Ω'' contains l vertical and k horizontal rows, and Ω''' contains l vertical and m horizontal rows. From any m of the l vertical rows of Ω''' form a determinant, and call this Q'''; from the $l - m$ supplementary vertical rows of Ω'', choosing the horizontal rows at pleasure, form a determinant, and call this Q''; from the $k - l + m$ supplementary horizontal rows of Ω', choosing the vertical rows at pleasure, form a determinant, and call this Q'; from the $h - k + l - m$ supplementary vertical rows of Ω form a determinant, and call this Q: then Q''' divides Q'', this quotient divides Q', this quotient divides Q, and $\nabla = Q \div \{Q' \div (Q'' \div Q''')\}$. The mode of proceeding is obvious.

It is clear, that if all the coefficients a, β, ... be considered of the order unity, ∇ is of the order $h - 2k + 3l - \cdots$.

What has preceded constitutes the theory of elimination alluded to in my memoir "On the Theory of Involution in Geometry," *Journal*, vol. II. p. 52–61, [40]. And thus the problem of eliminating any number of variables x, y, ... from the same number of equations $U = 0$, $V = 0$, ... (where U, V, ... are homogeneous functions of any orders whatever) is completely solved; though, as before remarked, I am not in possession of any method of arriving at once at the final result in its most simplified form; my process, on the contrary, leads me to a result encumbered by an extraneous factor, which is only got rid of by a number of successive divisions less by two than the number of variables to be eliminated.

To illustrate the preceding method, consider the three equations of the second order,

$$U = ax^2 + by^2 + cz^2 + lyz + mzx + nxy = 0,$$

$$V = a'x^2 + b'y^2 + c'z^2 + l'yz + m'zx + n'xy = 0,$$

$$W = a''x^2 + b''y^2 + c''z^2 + l''yz + m''zx + n''xy = 0.$$

Here, to eliminate the fifteen quantities x^4, y^4, z^4, y^3z, z^3x, x^3y, yz^3, zx^3, xy^3, y^2z^2, z^2x^2, x^2y^2, x^2yz, y^2zx, z^2xy, we have the eighteen equations

$$x^2U = 0, \quad y^2U = 0, \quad z^2U = 0, \quad yzU = 0, \quad zxU = 0, \quad xyU = 0,$$

$$
\begin{gathered}
x^2V=0,\quad y^2V=0,\quad z^2V=0,\quad yzV=0,\quad zxV=0,\quad xyV=0,\\
x^2W=0,\quad y^2W=0,\quad z^2W=0,\quad yzW=0,\quad zxW=0,\quad xyW=0,
\end{gathered}
$$

equations, however, which are not independent, but are connected by

$$
\begin{aligned}
&a''x^2V+b''y^2V+c''z^2V+l''yzV+m''zxV+n''xyV\\
&\qquad-(a'x^2W+b'y^2W+c'z^2W+l'yzW+m'zxW+n'xyW)=0,
\end{aligned}
$$

$$
\begin{aligned}
&ax^2W+by^2W+cz^2W+lyzW+mzxW+nxyW\\
&\qquad-(a''x^2U+b''y^2U+c''z^2U+l''yzU+m''zxU+n''xyU)=0,
\end{aligned}
$$

$$
\begin{aligned}
&a'x^2U+b'y^2U+c'z^2U+l'yzU+m'zxU+n'xyU\\
&\qquad-(ax^2V+by^2V+cz^2V+lyzV+mzxV+nxyV)=0.
\end{aligned}
$$

Arranging these coefficients in the required form, we have the following value of ∇.

$$
\nabla=\left|\begin{array}{cccccc|cccccc|cccccc}
a&&&&&&a'&&&&&&a''&&&&&\\
&b&&&&&&b'&&&&&&b''&&&&\\
&&c&&&&&&c'&&&&&&c''&&&\\
&l&&b&&&&l'&&b'&&&&l''&&b''&&\\
&&m&&c&&&&m'&&c'&&&&m''&&c''&\\
n&&&&&a&n'&&&&&a'&n''&&&&&a''\\
&&l&c&&&&&l'&c'&&&&&l''&c''&&\\
m&&&&a&&m'&&&&a'&&m''&&&&a''&\\
&n&&&&b&&n'&&&&b'&&n''&&&&b''\\
&c&b&l&&&&c'&b'&l'&&&&c''&b''&l''&&\\
c&&a&&m&&c'&&a'&&m'&&c''&&a''&&m''&\\
b&a&&&&n&b'&a'&&&&n'&b''&a''&&&&n''\\
l&&&a&n&m&l'&&&a'&n'&m'&l''&&&a''&n''&m''\\
&m&&n&b&l&&m'&&n'&b'&l'&&m''&&n''&b''&l''\\
&&n&m&l&c&&&n'&m'&l'&c'&&&n''&m''&l''&c''\\
&&&&&&a''&b''&c''&l''&m''&n''&-a'&-b'&-c'&-l'&-m'&-n'\\
-a''&-b''&-c''&-l''&-m''&-n''&&&&&&&a&b&c&l&m&n\\
a'&b'&c'&l'&m'&n'&-a&-b&-c&-l&-m&-n&&&&&&
\end{array}\right|
$$

which may be represented as before by

$$
\nabla=\left|\frac{\Omega}{\Omega'}\right|.
$$

Thus, for instance, selecting the first, second, and sixth lines of Ω' to form the determinant Q', we have $Q'=a''(a'b''-a''b')$; and then Q must be formed from the third, fourth, fifth, seventh, &c. ... eighteenth lines of Ω. (It is obvious that if Q' had been formed from the first, second, and third lines of Ω', we should have had $Q'=0$; the corresponding value of Q would also have vanished, and an illusory result be obtained; and similarly for several other combinations of lines.)

60.

ON THE EXPANSION OF INTEGRAL FUNCTIONS IN A SERIES OF LAPLACE'S COEFFICIENTS.

[From the Cambridge and Dublin Mathematical Journal, vol. iii. (1848), pp. 120, 121.]

Suppose

$$S = A\mu^s + A_1\mu^{s-1} + \dots,\\ \quad = a_0 Q_s + a_1 Q_{s-1} + \dots, \tag{1}$$

where, as usual, Q_0, Q_1, &c. are the coefficients of the successive powers of p in $(1-2\mu p+p^2)^{-\frac{1}{2}}$. Assume

$$S = \frac{\sqrt{(1+\Delta^2)}}{\sqrt{(1-2\mu\Delta+\Delta^2)}}\,C, \tag{2}$$

where Δ refers to C; then expanding this expression, first in powers of μ, and then in a series of terms of the form $\sqrt{(1+\Delta^2)}\,Q$, and comparing these with the preceding values of S,

$$A_{s-q} = \frac{1\cdot 3\cdots(2q-1)}{2\cdot 4\cdots 2q}\left(\frac{2\Delta}{1+\Delta^2}\right)^q C,$$

$$a_{s-r} = \sqrt{(1+\Delta^2)}\,\Delta^r C. \tag{3}$$

Assume $\dfrac{2\Delta}{1+\Delta^2} = \delta$, that is,

$$\Delta = \frac{1}{\delta}\{1-\sqrt{(1-\delta^2)}\}, \quad \sqrt{(1+\Delta^2)} = \frac{\sqrt{2}}{\delta}\{1-\sqrt{(1-\delta^2)}\}^{\frac{1}{2}}. \tag{4}$$

Then

$$\delta^q C = \frac{2\cdot 4\cdots 2q}{1\cdot 3\cdots(2q-1)}A_{s-q},$$

$$a_{s-r} = \frac{\sqrt{2}}{\delta^{r+1}}\{1-\sqrt{(1-\delta^2)}\}^{r+\frac{1}{2}}C. \tag{5}$$

or, expanding this last equation in powers of δ,

$$a_{s-r} = \frac{(2r+1)}{2^r}\delta^r\left(\frac{1}{2r+1}+\frac{1}{2}\frac{\delta^2}{2^2}+\frac{(2r+7)}{2\cdot 4}\frac{\delta^4}{2^4}+\frac{(2r+9)(2r+11)}{2\cdot 4\cdot 6}\frac{\delta^6}{2^6}+\dots\right)C. \tag{6}$$

or, replacing the successive terms of the form $\delta^q\cdot C$ by their respective values,

$$a_{s-r} = (2r+1)\left\{\frac{2\cdot 4\cdots 2r}{3\cdot 5\cdots(2r+1)2^r}A_{s-r}+\frac{4\cdot 6\cdots(2r+4)}{3\cdot 5\cdots(2r+3)2^{r+2}}A_{s-r-2}+\dots\right.\\ \left. +\frac{(2k+2)(2k+4)\cdots(2r+4k)}{3\cdot 5\cdots(2r+2k+1)2^{r+2k}}A_{s-r-2k}+\dots\right\}. \tag{7}$$

Thus, if $S=\mu^s$, so that $A_{s-r}=0$, except in the particular case $A=1$,

$$a_{2k-1} = 0,$$

$$a_{2k} = \frac{2s-4k+1}{2^s}\,\frac{(2k+2)(2k+4)\cdots 2s}{3\cdot 5\cdots(2s-2k+1)}. \tag{8}$$

or

$$\mu^s = \frac{1}{2^s}\Sigma\left\{(2s-4k+1)\frac{(2k+2)(2k+4)\cdots 2s}{3\cdot 5\cdots(2s-2k+1)}Q_{s-2k}\right\}, \tag{9}$$

which of course includes the preceding case. By substituting the expanded values of the coefficients Q, or again, by determining the value of $(1-\mu)^s$ in terms of these coefficients, and equating it with that given in Murphy's *Electricity*, [8°. Cambridge, 1833], p. 10, or in a variety of other ways, a series of identical equations involving sums of factorials may readily be obtained. The mode of employing the general theory of the separation of symbols made use of in the preceding example, may easily be applied to the solution of analogous questions.

61.

ON GEOMETRICAL RECIPROCITY.

[From the Cambridge and Dublin Mathematical Journal, vol. iii. (1848), pp. 173–179.]

The fundamental theorem of reciprocity in plane geometry may be thus stated.

"The points and lines of a plane P may be considered as corresponding to the lines and points of a plane P' in such a manner that to a set of points in a line in the first figure, there corresponds a set of lines through a point in the second figure, (namely through the point corresponding to the line); and to a set of lines through a point in the first figure, there corresponds a set of points in a line in the second figure, (namely in the line corresponding to the point)."

And from this theorem, without its being in any respect necessary further to particularize the nature of the correspondence, or to consider in any manner the relative position of the two planes, an endless variety of propositions and theories may be deduced, as, for instance, the duality of all theorems which relate to the purely descriptive properties of figures, the theory of the singular points and tangents of curves, &c.

Suppose, however, that the two planes coincide, so that a point may be considered indifferently as belonging to the first or to the second figure: an entirely independent series of propositions (which, properly speaking, form no part of the general theory of reciprocity) result from this particularization. In general, the line in the second figure which corresponds to a point considered as belonging to the first figure, and the line in the first figure which corresponds to the same point considered as belonging to the second figure, will not be identical; neither will the point in the second figure which corresponds with a line considered as belonging to the first figure, and the point in the first figure which corresponds to the same line considered as belonging to the second figure, be identical.

In the particular case where these lines and points are respectively identical (the identity of the lines implies that of the points and vice versâ) we have the theory of "reciprocal polars." Here, where it is unnecessary to define whether the points or lines belong to the first or second figures, the line corresponding to a point and the point corresponding to a line are spoken of as the polar of the point and the pole of the line, or as reciprocal polars.

"The points which lie in their respective polars are situated in a conic, to which the polars are tangents." Or, stating the same theorem conversely,

"The lines which pass through their respective poles are tangents to a conic, the points of contact being the poles."

To determine the polar of a point, let two tangents be drawn through this point to the conic, the points of contact are the poles of the tangents; hence the line joining them is the polar of the point of intersection of the tangents, that is, "The polar of a point is the line joining the points of contact of the tangents which pass through the point."

Conversely, and by the same reasoning,

"The pole of a line is the intersection of the tangents at the points where the line meets the conic."

The actual geometrical constructions in the several cases where the point is within or without the conic, or the line does or does not intersect the conic, do not enter into the plan of the present memoir.

Passing to the general case where the lines and points in question are not identical, which I should propose to term the theory of "Skew Polars" (Polaires Gauches), we have the theorem,

"Considering the points in the first figure which are situated in their respective corresponding lines in the second figure, or the points in the second figure which are situated in their respective corresponding lines in the first figure, in either case the points are situated in the same conic (which will be spoken of as the 'pole conic'), and the lines are tangents to the same conic (which will be spoken of as the 'polar conic'), and these two conics have a double contact." This theorem is evidently identical with the converse theorem.

The corresponding lines to a point in the pole conic are the tangents through this point to the polar conic; viz. one of these tangents is the corresponding line when the point is considered as belonging to the first figure, and the other tangent is the corresponding line when the point is considered as belonging to the second figure.

The corresponding points to a tangent of the polar conic are the points where this line intersects the pole conic; viz. one of these points is the corresponding point when the line is considered as belonging to the first figure, and the other is the corresponding point when the line is considered as belonging to the second figure.